

Java Exception Handling Report

Experiment: StringIndexOutOfBoundsException

1. What is StringIndexOutOfBoundsException?

StringIndexOutOfBoundsException occurs when a program tries to access a character in a String using an invalid index. Java String indexes start from **0** and go up to **length - 1**. In this program, the string **"Java"** has 4 characters, so valid indexes are 0, 1, 2, and 3. Trying to access index 10 causes the exception. The exception is caught and a meaningful message is displayed instead of terminating the program.

Note: This is different from NumberFormatException. NumberFormatException occurs when a String cannot be converted into a valid number, such as Integer.parseInt("ABC").

2. Objective / Aim

To write and execute a Java program that demonstrates StringIndexOutOfBoundsException by accessing an invalid String index and handling the exception using try-catch.

3. Task Description

Create a Java class named **StringIndexDemo**. Store the String "Java" in a variable, attempt to access character index 10 using **charAt()**, catch the resulting **StringIndexOutOfBoundsException**, and display an appropriate exception message.

4. Step-wise Execution

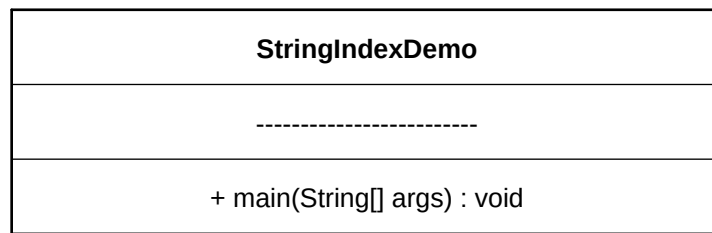
1. Start the program.
2. Create a String variable named **text** and assign it the value "Java".
3. The String contains 4 characters, so its valid indexes are 0 to 3.
4. Call **text.charAt(10)** to access the character at index 10.
5. Index 10 is invalid because it is outside the valid range 0 to 3.
6. Java throws StringIndexOutOfBoundsException.
7. The catch block catches the exception.
8. Print "Exception: String index is out of bounds".
9. End the program.

5. Algorithm

1. START
2. Declare String text = "Java".
3. Try to execute text.charAt(10).
4. If the index is invalid, an exception is generated.
5. Catch StringIndexOutOfBoundsException.
6. Display the exception message.

7. STOP

6. UML Class Diagram



Flow: main() → try block → charAt(10) → StringIndexOutOfBoundsException → catch block → output.

7. Program Code

```
class StringIndexDemo {
    public static void main(String[] args) {
        try {
            String text = "Java";
            System.out.println(text.charAt(10));
        } catch (StringIndexOutOfBoundsException e) {
            System.out.println("Exception: String index is out of bounds");
        }
    }
}
```

8. Code Explanation

- **class StringIndexDemo** – Defines the Java class.
- **main()** – The execution of the program begins here.
- **try** – Contains code that may generate an exception.
- **String text = "Java";** – Creates a String containing four characters.
- **text.charAt(10)** – Attempts to access the character at index 10. This is invalid because the maximum valid index is 3.
- **catch (StringIndexOutOfBoundsException e)** – Catches the exception generated by the invalid String index.
- **System.out.println(...)** – Displays a user-friendly message explaining the error.

9. Output

The program does not print a character because index 10 is invalid. The catch block handles the exception and prints:

```
Exception: String index is out of bounds
```

10. Output Screenshot

Command Prompt / Console Output

```
C:\> javac StringIndexDemo.java
C:\> java StringIndexDemo

Exception: String index is out of bounds
```